

Authorship Attribution and Style Fingerprinting in Bangla Literature

Project Proposal

1. Overview and Motivation

Authorship attribution asks: given an unseen passage, which author (from a closed candidate set) most likely wrote it, based on *how* something is written rather than *what* it says. Bangla literature spans stylistically distinct classical and modern voices, yet computational stylometry for Bangla remains comparatively underexplored, and per-author corpora are typically small. This project builds a multi-author Bangla stylometric pipeline that combines interpretable stylometric features with word embeddings, and directly compares a **generative** attribution paradigm against a **discriminative** one — while also asking a representation-level question: in a small-corpus regime, do **pretrained** embeddings or embeddings **trained from scratch** on the author corpus better capture stylistic signal rather than general semantics?

2. Objectives

- Curate and document a cleaned, public-domain, multi-author Bangla corpus.
- Extract stylometric features: function-word frequencies, sentence-length distributions, POS n -gram statistics.
- Compare pretrained vs. from-scratch embeddings for isolating stylistic signal from semantic content.
- Implement and compare a generative per-author language-model attributor against a discriminative SVM / fine-tuned BERT classifier.
- Analyze confusion between stylistically similar authors and interpret the features driving each model's predictions.
- Ship a simple interface: paste a passage, receive the predicted author and the top features behind the decision.

3. Methodology

3.1 Corpus construction

Public-domain works are collected across classical and modern Bangla authors, with sentence segmentation, normalization, deduplication, and removal of front/back matter. Sampling is balanced per author to limit corpus-size skew, and the full collection-and-cleaning process is documented as part of the methodology, not treated as a black box.

3.2 Feature engineering

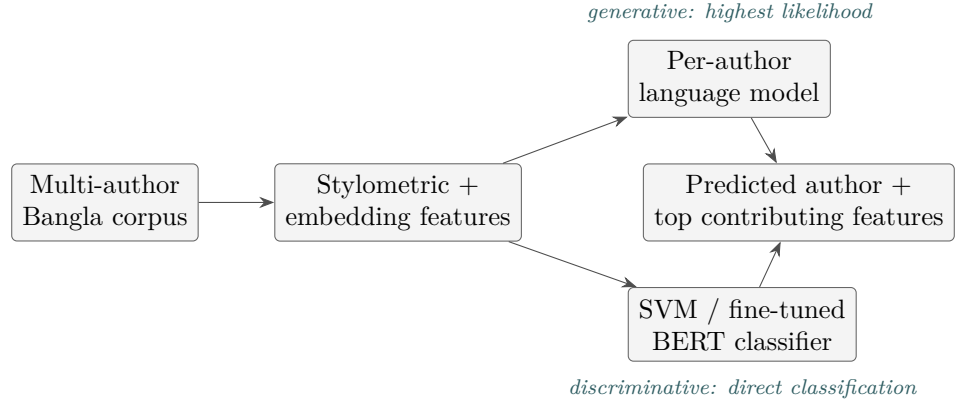
Stylometric: function-word frequency vectors (pronouns, postpositions, particles), sentence-length distribution statistics (mean, variance, skew), and POS n -grams from a Bangla tagger.

Embeddings: (a) pretrained Bangla embeddings (e.g. fastText / BanglaBERT) and (b) embeddings trained from scratch on the author corpus, evaluated on how well each isolates author style versus topical content.

3.3 Two attribution paradigms

Generative: a per-author language model is fit on each author's text; an unseen passage is attributed to the author whose model assigns it the highest likelihood (lowest perplexity).

Discriminative: stylometric and embedding features are concatenated and fed to an SVM, and separately used to fine-tune a BERT classifier; both are trained directly to discriminate between authors.



3.4 Worked example

Input passage (constructed example): বৃষ্টি থামলে সে জানালার পাশে দাঁড়িয়ে দূরের আকাশের দিকে তাকিয়ে রইল।

Extracted signal: sentence length = 12 words · function words পাশে, দিকে (postpositions), সে (pronoun) · POS pattern NOUN--VERB--ADP--NOUN

Generative model: log-likelihood highest under Author B's language model.

Discriminative model: SVM assigns Author B probability 0.71, driven mostly by postposition frequency and mean sentence length.

Output: *Predicted author:* Author B — top features: postposition rate, mean sentence length, NOUN--VERB--ADP rate.

4. Evaluation Plan

- **Metrics:** accuracy and macro-F1 (per-author class imbalance), with confusion-matrix analysis focused on stylistically close authors.
- **Ablations:** stylometric-only vs. embedding-only vs. combined features; pretrained vs. from-scratch embeddings.
- **Interpretability:** feature-importance / SHAP for the SVM; likelihood-contribution breakdown for the generative models; comparison of which feature families drive each paradigm's correct and incorrect predictions.

5. Interface

A minimal input–output interface accepts a pasted Bangla passage and returns the predicted author alongside the ranked stylistic features that most influenced the decision, so predictions remain auditable rather than opaque.

6. Expected Outcomes

A documented, reusable multi-author Bangla stylometric corpus; an empirical comparison of generative and discriminative attribution for a morphologically rich, low-resource language; and evidence on whether pretrained or from-scratch embeddings better isolate authorial style in small-corpus settings.